

RESUME

Abusufiyan

Chemical Engineer | Research Associate - Automation Engineer
Hyderabad, India

Phone: +91 9845345067

Email: ptlabusufiyanpatel@gmail.com

LinkedIn: [linkedin.com/in/sufiyanabu](https://www.linkedin.com/in/sufiyanabu)

Website: labbuddy.org



Professional Summary

Chemical Engineer with around 2 years of experience in pharmaceutical R&D support, with hands-on work in continuous flow chemistry, process automation, laboratory instrumentation, and smart laboratory systems. Experience includes operating and automating Vapourtec flow reactors, fixed-bed hydrogenation systems, HITEC ZANG LabManager workflows, and custom digital control solutions for research environments. Inventor on a filed Indian patent related to laboratory device control, with strong interest in bridging chemical engineering, process development, and automation for safer and more efficient pharmaceutical operations.

Core Competencies

- Continuous flow chemistry
- Process development support
- Fixed-bed hydrogenation
- Laboratory automation
- Process instrumentation and control
- HITEC ZANG LabManager
- Data acquisition and safety interlocks
- Embedded systems and PCB development
- RS-232 / RS-485 device integration
- SOP preparation and workflow standardization

Professional Experience

Dr. Reddy's Institute of Life Sciences, Hyderabad

Research Associate - Automation Engineer

May 2024 - Present

- Working within the 10X Lab of Excellence to support pharmaceutical R&D through continuous flow chemistry, process development, laboratory automation, and instrumentation engineering.
- Operate and support Vapourtec R-Series continuous-flow platforms, HPLC pumps, peristaltic pumps, and multiple reactor configurations for reaction development and experimental workflow execution.Profile.
- Collaborate with research chemists to configure flow rate, reactor temperature, and residence time during continuous-flow experiments and process evaluation studies.
- Support fixed-bed catalytic hydrogenation studies using catalyst-packed reactors, liquid delivery systems, and mass flow controllers under controlled hydrogen handling conditions.Profile.
- Conduct solvent screening and solubility studies to assess suitability of reactions for continuous-flow processing and laboratory execution.
- Configure and customize HITEC ZANG LabManager for coordinated operation of pumps, reactors, sensors, and associated laboratory equipment in integrated experimental workflows.Profile.
- Develop automation solutions for standalone and legacy laboratory instruments, including RS-232 and RS-485 communication-based integration for centralized monitoring and control.
- Design and integrate instrumentation systems involving PT100 sensors, PID controllers, DAQ systems, pH sensors, stirrers, pumps, and safety interlocks for real-time process monitoring and improved operational reliability.Profile.
- Build custom embedded hardware, PCB-based monitoring devices, and equipment control interfaces to retrofit conventional laboratory systems into digitally controlled and data-enabled platforms.Profile.
- Prepare SOPs, support onboarding activities, and contribute to improving repeatability and standardization in automated laboratory operations.
- Led the technical development of a laboratory equipment interface platform that formed the basis of the filed Indian patent titled A System for Controlling an External Device and Method Thereof.Profile.

Selected Projects

- **LabBuddy.org** “*Integrated Laboratory Knowledge Platform*”
 - Independently designed and developed a laboratory knowledge platform that consolidates scientific resources commonly used in pharmaceutical R&D into a single searchable interface.
 - Provides quick access to chemical and API information, safety resources, GHS data, analytical references, and laboratory calculators to support day-to-day research activities.
 - Created to reduce time spent searching across multiple databases while improving accessibility of reliable scientific information.
 - Continuously enhanced with new tools and scientific datasets based on practical laboratory workflows.
- **Protocol Bridge for Legacy RS-232/RS-485 Devices:** Developed a configurable communication layer and LAN-based interface to connect legacy laboratory instruments into a centralized digital workflow.
- **Networked Syringe Pump:** Designed a standalone syringe pump with remote monitoring, data logging, and future suitability for continuous-flow applications.
- **Distributed PT100 Temperature Monitoring System:** Built a modular monitoring architecture focused on reliability, scalability, and reduction of single-point failures.
- **Digital Retrofitting of Analog Power Modules:** Created a non-invasive digital control interface for legacy analog power hardware while preserving original circuitry.

Patent

Inventor - “*A System for Controlling an External Device and Method Thereof*”

Indian Patent Application No. **IN202541044701**

Filed on 8 May 2025.

Technical Skills

Process Development: Continuous flow chemistry, hydrogenation, solvent screening, process safety, reaction system support.

Laboratory Automation: HITEC ZANG LabManager, equipment integration, experimental workflow automation, SOP development, laboratory digitalization.

Instrumentation: PT100 sensors, PID controllers, DAQ systems, MFCs, HPLC pumps, peristaltic pumps, syringe pumps, diaphragm pumps, pH sensors, fixed-bed reactors, microreactors, plug flow reactors, Corning and Vapourtec flow reactors.

Embedded and Engineering Tools: KiCad, STM32, ESP32, Arduino, PCB design, firmware basics, Raspberry Pi, Fusion 360, 3D printing.

Software: Python, MATLAB, LabVIEW, Power BI, Microsoft Office, Linux.

Education

Sl.no	Examination	Board/University	year of passing	Percentage/ CGPA
01.	Bachelor of Engineering (Chemical Engineering)	Visvesvaraya Technological University	2023	6.4
02.	Per University (12)	Dept. of the pre-university education broad	2019	75.83%
03.	S.S.L.C examination	Karnataka Secondary Education Examination Board	2017	80.16%

Professional Reference

Maheshkumar Uppada

Principal Research Scientist & Head of AI Driven Process Technologies - 10X Lab of Excellence

Dr. Reddy's Institute of Life Sciences

Email: maheshkumaru@drils.org

Relationship: Reporting Manager.

Biodata

Field	Details
Full Name	Abu Sufiyan
Father's Name	Dr.Ibrahim Patel
Date of Birth	01/12/2001
Gender	Male
Languages Known	English, Hindi, Kannada, Marathi
Current Address	Hyderabad, Telangana
Permanent Address	2-105-4, Cb colony road,Sahu Nagar, Bhalki, Karnataka 585328, India
Mobile	+91 9845345067
Email	ptlabusufiyanpatel@gmail.com